

MATH 341: Homework 4

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1 Question 1

What lower triangular matrix E puts A into upper triangular form $EA = U$?
Multiply by $E^{-1} = L$ to factor A into LU :

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix}$$

1.1 Finding E

$$\underbrace{\begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix}}_A \xrightarrow{R_3=R_3-3R_1} \underbrace{\begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix}}_U \Rightarrow \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}}_E$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}}_E \xrightarrow{\text{Multiply off-diag } -1} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}}_L$$

2 Question 2

This problem shows how the one-step inverses multiply to give L . You see this best when $A = L$ is already lower triangular with 1's on the diagonal. Then $U = I$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix} \quad E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ -b & 0 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -c & 1 \end{bmatrix}$$

- E_2E_1 to find $EA = I$
- $E_1^{-1}E_2^{-1}$ to find $A = L$
- The multipliers a, b, c are mixed up in $E = L^{-1}$ but are perfect in L

3 Question 3

Compute L, U for the matrix A :

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$$

Find four condition on a, b, c, d to get $A = LU$ with four non-zero pivots.

4 Question 4

What multiple of a should be subtracted from b to make the result A_2 orthogonal to a ? Sketch a figure to show a, b, A_2

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

5 Question 5

Complete the Gram-Schmidt process in the prior problem via computing

$$a_1 = \frac{a}{\|a\|} \quad A_2 = b - (b^T q_1)q_1 \quad q_2 = \frac{A_2}{\|A_2\|}$$